CHEMICAL KINETICS

1. For the reaction–

$$5Br^-(aq) + BrO_3^-(aq) + 6H^{\oplus}(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$$

Rate of disappearance of $Br^-(aq)$ is

1.5×10^{-2} mol L^{-1} sec^{-1}.

Rate of appearance of $Br_2(aq)$ is–

(1) 3×10^{-3} mol L^{-1} s^{-1}

(2) 6×10^{-2} mol L^{-1} s^{-1}

(3) 9×10^{-3} mol L^{-1} s^{-1}

(4) 4.5×10^{-2} mol L^{-1} s^{-1}

2. The decomposition of N_2O_5 in CCl_4 at 318K has been studied by monitoring the concentration of N_2O_5 in the solution. Initially the concentration of N_2O_5 is 3.32 mol L^{-1} and after 120 minutes, it is reduced to 1.04 mol L^{-1}. The reaction takes place according to the equation–

$$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

What is the rate of production of $O_2(g)$

(1) 1.14 mol L^{-1} h^{-1} (2) 0.57 mol L^{-} h^{-1}

(3) 2.28 mol L^{-1} h^{-1} (4) None of these

3. A chemical reaction $2A \longrightarrow 4B + C$ in gaseous phase shows an increase in concentration of B by 5×10^{-3} M in 10 second. The rate of reaction and rate of appearance of B are respectively–

(1) 5×10^{-4} Ms^{-1} and 1.25×10^{-4} Ms^{-1}

(2) 1.25×10^{-4} Ms^{-1} and 5×10^{-4} Ms^{-1}

(3) 1.25×10^{-4} Ms^{-1} and 2.5×10^{-4} Ms^{-1}

(4) 2.5×10^{-4} Ms^{-1} and 1.25×10^{-4} Ms^{-1}

4. For the decomposition,

$$N_2O_5(g) \longrightarrow N_2O_4(g) + \frac{1}{2}O_2(g),$$

the initial pressure of N_2O_5 is 114 mm and after 20 sec, the pressure of reaction mixture becomes 133 mm. The rate of reaction in terms of atm sec^{-1} is–

(1) 1.25×10^{-3} atm sec^{-1}

(2) 2.5×10^{-3} atm sec^{-1}

(3) 9.5×10^{-1} atm sec^{-1}

(4) 4.75×10^{-1} atm sec^{-1}

5. Which of the following is correct–

(1) Decomposition of NH_3 on finely divided platinum surface is first order when the pressure of NH_3 is low

(2) Decomposition of NH_3 on finely divided platinum surface is zero order. when the presure of NH_3 is high

(3) The thermal decomposition of HI on gold surface is zero order reaction.

(4) All of these

6. The rate constant of first order reaction becomes 6 times when the temperature is increased from 350 K to 410 K. Energy of activation for the reaction is–

(1) 35.63 KJ mol^{-1} (2) 45.63 KJ mol^{-1}

(3) 25.20 KJ mol^{-1} (4) 23.62 KJ mol^{-1}

7. For the reaction, $2A + B + C \longrightarrow A_2B + C$, the rate law has been determined to be

Rate = K [A] $[B]^2$ $[C]^0$

If the value of K is 2.0×10^{-6} mol^{-2} L^2 s^{-1} for the reaction, Initial rate of the reaction with

(A) = 0.2 mol L^{-1} ;

(B) = 0.1 mol L^{-1} ;

(C) = 0.5 mol L^{-1} ;

is –

(1) 2×10^{-9} Ms^{-1} (2) 4×10^{-9} Ms^{-1}

(3) 3×10^{-9} Ms^{-1} (4) 1×10^{-9} Ms^{-1}

8. Rate constant for first order reaction is 5.78×10^{-5} s^{-1}. What percentage of initial reactant will react in 10 hours?

(1) 12.5% (2) 25% (3) 87.5% (4) 75%

9. A first order reaction with respect to reactant A has a rate constant 6.0×10^{-2} min^{-1}. If we start with [A] = 0.5 mol L^{-1}; when would (A) reach the value 0.05 mol L^{-1}?

(1) 48.4 min (2) 28.9 min

(3) 38.4 min (4) 18.6 min

10. In a reaction $2A \longrightarrow$ products, the concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes, rate of reaction is–

(1) 0.005 mol L^{-1} min^{-1}

(2) 0.002 mol L^{-1} min^{-1}

(3) 0.05 mol L^{-1} min^{-1}

(4) 0.02 mol L^{-1} min^{-1}

11. For a first order reaction half life periods are 5000 and 1000 seconds at300 K and 310 K temperature respectively. Energy of activation is–

(1) 12.46 KJ mol^{-1} (2) 124.46 KJ mol^{-1}

(3) 46.12 KJ mol^{-1} (4) 12.46 KJ mol^{-1}

12. The experimental data for the reaction $2A + B_2 \longrightarrow 2AB$ are as follows. Rate law expression will be–

[A] mol L^{-1}	[B_2] mol L^{-1}	Rate mol L^{-1} sec^{-1}
0.50	0.50	1.6×10^{-4}
0.50	1.0	3.2×10^{-4}
1.00	1.0	3.2×10^{-4}

(1) rate = $K[B_2]^1$ (2) rate = $K[A]^1[B_2]^1$

(3) rate = $K[A]^1$ (4) rate = $K[A]\,[B_2]^2$

13. The half life period and initial concentration for a reaction are as follows:

Initial concentration	350	540	158
t ½	425	275	941

order of reaction is–

(1) 2 (2) 1

(3) 0 (4) 4

14. The activation energy for the reaction–

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

is 18 K cal/mol at 300 K. calculate the fraction of molecules of reactants having energy equal to or greater than activation energy?

Anti log (–13.02) = 9.36×10^{-14}

(1) 9.36×10^{-14} (2) 1.2×10^{-12}

(3) 4.2×10^{-16} (4) 5.2×10^{-15}

15. Which of the following is false about catalyst–

(1) A catalyst does not alter Gibb's energy; ΔG of a reaction.

(2) It catalyses the non-spontaneous reactions

(3) Catalyst does not change the equilibrium constant of a reaction.

(4) A small amount of the catalyst can catalyse a large amount of reactants.

16. For a first order homogeneous gaseous reaction, $A \longrightarrow 2B + C$, initial pressure was P_i while total pressure after time 't' was 'P_t'. The write expression for the rate constant K in terms of P_i, P_t and t is–

(1) $K = \frac{2.303}{t}\log\left(\frac{2P_i}{3P_i - P_t}\right)$

(2) $K = \frac{2.303}{t}\log\left(\frac{2P_i}{2P_t - P_i}\right)$

(3) $K = \frac{2.303}{t}\log\left(\frac{P_i}{P_i - P_t}\right)$

(4) None of these

17. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 minutes at 47°C. The energy of activation of the reaction is–

(1) 43.85 KJ (2) 55.33 KJ

(3) 11.97 KJ (4) 6.65 KJ

18. Half life period of U^{237} is 2.5×10^5 years. In how much time will the amount of U^{237} remained 25% of the original amount?

(1) 2.5×10^5 years (2) 1.25×10^5 years

(3) 5×10^5 years (4) 10^6 years

19. The plot of lnk versus 1/T is linear with slope of–

(1) $\frac{-E_a}{R}$ (2) $\frac{E_a}{R}$

(3) $\frac{E_a}{2.303R}$ (4) $\frac{-E_a}{2.303R}$